

BioSpecInfo

Traceability Matrix — BioSpecInfo

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Traceability Matrix — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Version described	bsi-v189
Purpose	Connect every declared requirement to the implementation that realises it and to the bench that verifies it.

How to read this matrix. Every row is a closed chain: a requirement, the point in the code that realises it, and the bench that verifies it works. A row without a bench is an **unverified** requirement and is declared as such, not hidden: the usefulness of a matrix lies precisely in making the gaps visible.

The outcomes of the benches cited here are reported in full, with their complete output, in [../evidence/RAPPORTO-VERIFICA.md](#), which is generated automatically.

1. Scientific requirements

ID	Requirement	Implementation	Verification bench
SCI-01	A drug structure must come from an external source and match that source, not merely weigh correctly	<code>FARM_DATA</code> in <code>index.html</code> — 263 entries , 36 declaring a ChEMBL identifier and formula	<code>audit_farmaci</code> — the formula is rebuilt by counting the graph's atoms and compared with ChEMBL's: 36 of 36. Weight alone was not enough: three pre-existing entries (lenalidomide, palbociclib, aripiprazole) were isomers — same weight, different molecule — and were replaced. Two ratchets: verified provenance $\geq$ 36, readable structures $\geq$ 240
SCI-02	Functional groups must be recognised on the structure, not on the text of the SMILES	<code>bsi-spettri.js</code> §1, SMARTS patterns via RDKit	<code>test_spettri</code> — 8 reference molecules, additive recognition
SCI-03	A molecule with several functional groups must show them all	<code>bandeDaGruppi()</code> , additive construction	<code>test_spettri</code> §1 — acetylsalicylic acid: ester and acid

ID	Requirement	Implementation	Verification bench
SCI-04	Spectra must respect the conventions of their technique	<code>svgIR()</code> , <code>makeNMRsvg()</code> , <code>drawUVSpectrum()</code> , <code>drawMSSpectrum()</code>	<code>test_assi</code> (SVG), <code>test_assi_canvas</code> (canvas)
SCI-05	A computed spectrum must be reproducible	every <code>Math.random()</code> removed from the traces	<code>test_assi</code> §3, <code>test_spettri</code> §8 — byte-for-byte comparison
SCI-06	The height of ¹ H peaks must represent the integration, or not be asserted at all	<code>makeNMRsvg()</code> ; <code>_nmrSvg()</code> draws bands of uniform height	<code>test_assi</code> §4
SCI-07	Conjugation must act on the individual carbonyl	SMARTS <code>acidoConiug</code> , <code>estereConiug</code> , <code>chetoneConiug</code> , <code>aldeideConiug</code>	<code>test_spettri</code> §1 — aspirin 1760/1690; acetophenone 1690
SCI-08	An ambiguous physical constant must not be resolved arbitrarily	<code>costante_fisica</code> in <code>bsi-ai-hub.js</code> — collects every candidate	<code>test_costanti</code> — 45 checks, including the refusals for ambiguity
SCI-09	The limits of the predictors must be declared to the user	chart headers; <code>docs/06</code> §3.5	<code>verifica_guida</code> — consistency between promises and code
SCI-10	Molecular symmetry must determine the selection rules correctly	<code>ssimm</code> section in <code>index.html</code>	<code>test_simmetria</code> — 26 checks: normal modes, centre of inversion, mutual exclusion
SCI-12	A chromatogram must follow from the parameters, not be drawn by hand	<code>scroma</code> section: $\sigma = t \sqrt{R}$, $R = s$ from distance and widths	<code>audit_stabilita</code> (opening and drawing), inspection of the values against the definitions
SCI-13	A calibration line must declare the uncertainty of its coefficients and recognise the two ways it misleads	<code>staratura</code> section: ordinary least squares, F-test line → parabola, deleted residual	Verified on the three example sets included: linear ($F = 0.3$), non-linear ($F = 186.6 > 7.71$), with an outlier (1 detected)
SCI-14	A QSAR model must be evaluated on scaffolds never seen, not on analogues from the same set	<code>bsi-cheminfo.js</code> §9 — <code>divisionePerScaffold()</code> alongside <code>divisioneCasuale()</code>	<code>test_cheminfo</code> — the scaffold split leaves no scaffold shared between training and test
SCI-15	A model score must be compared against chance, not presented on its own	<code>bsi-cheminfo.js</code> §9 — retraining on scrambled labels, eight repetitions	<code>test_cheminfo</code> — verified in both directions : on a learnable signal R^2 0.394 beats all eight twins (margin 0.287); on random labels R^2 -0.199 does not beat them

ID	Requirement	Implementation	Verification bench
SCI-16	Two spellings of the same molecule must not be counted twice, nor land on opposite sides of the split	<code>bsi-cheminfo.js</code> §1 — deduplication on the canonical SMILES after the largest fragment	<code>test_cheminfo</code> — <code>OC(=O)C</code> and <code>CC(=O)O</code> collapse into one entry; salts lose the counterion
SCI-17	Similarity between molecules must be computed, not estimated	<code>bsi-cheminfo.js</code> §3 — Tanimoto and Dice over Morgan, RDKit, pattern and MACCS fingerprints	<code>test_cheminfo</code> — compared against hand-computable values, including the $0/0 = 1$ convention
SCI-18	A validation on a single split, over a small set, is a noisy number: it must come with its dispersion	<code>bsi-cheminfo.js</code> §9-bis — <code>validazioneIncrociata()</code> over scaffold-grouped folds	<code>test_cheminfo</code> — the folds cover every molecule once, no scaffold sits in two folds , and the same seed gives the same folds
SCI-19	A result that changes on every run is verifiable by nobody	seeded mulberry32 generator for splits, scrambles and folds; the seed is written into the report	<code>test_cheminfo</code> — two calls with the same seed give identical folds; the method report declares the seed
SCI-20	Results must be able to leave the tool, and the method must leave with them	<code>bsi-cheminfo.js</code> §9-ter — <code>esportaCsv()</code> and <code>rapportoMetodo()</code> , both formed in memory in the browser	<code>test_cheminfo</code> — columns consistent with the header, a name containing a comma quoted, residual computed, empty cell for missing activity; the report carries seed, fingerprint, verdict, limits
SCI-21	A number presented as structural similarity must come from a fingerprint, and one implementation must compute it	<code>rdkit_lab.html</code> loads <code>bsi-cheminfo.js</code> and uses its <code>tanimoto()</code> and <code>fingerprint()</code> ; the eight-bit similarity has been removed	<code>test_cheminfo</code> — page and engine give the same Tanimoto over all 28 pairs , deviation exactly 0 ; the fake-similarity functions no longer exist; every reference structure is valid and carries the monoisotopic mass of the drug it names
SCI-22	Conversions between molecular languages must be checked against an external source or against hand calculation, and coordinates written to a file must be real and declared	<code>bsi-molingue.js</code> — seventeen outputs , from SMILES to CML	<code>test_lingue</code> — 7 literature InChI keys , full SMILES → molfile → SMILES round-trip, 26 hand-written names, 7 required refusals ; mass composition against hand calculation (alanine C 40.44%, aspirin C 60.00%); XYZ and PDB with non-null coordinates and declared 2D

ID	Requirement	Implementation	Verification bench
SCI-23	Angles, lengths and dihedrals are measured on the coordinates, and on flat coordinates they are not shown	dot product for angles, cross product for dihedrals	<code>test_mol3d</code> — tetrahedron 109.4712° , water 104.47°, CO ₂ 180°, BF ₃ 120°, ammonia 106.13°; dihedrals 0°, 60°, 90°, 180°; ethanol C-C 1.509 Å; and the refusal tested in both directions
SCI-24	A spectrum read from a file must be decoded according to the format, and the peaks found must be the ones that are there — no more, no fewer	<code>bsi-spettrolettore.js</code> — JCAMP-DX with full ASDf (PMAI, DIF, DUP) and the integrity check the format prescribes; peaks by prominence , noise from MAD × 1.4826	<code>test_spettrolettore</code> — 39 checks . On three Gaussians built at 1715, 2950, 3400 cm ⁻¹ under noise it finds three: 1716, 2952, 3402. Both ways : on pure noise it finds zero . ASDf verified symbol by symbol (<code>abcdefghi</code> = -1...-9; <code>n</code> = -5, not -4)
SCI-25	A band must be declared <i>compatible with</i> , never assigned to a single possibility	<code>BANDE_IR</code> — 24 ranges; <code>assegnaIR()</code> returns all compatible assignments	<code>test_spettrolettore</code> — at 1715 cm ⁻¹ both a ketone C=O and a carboxylic acid C=O appear; the section states that it does not deduce the structure
SCI-26	A published model must be checked against its own declared error, and a learned model against chance	<code>bsi-cheminfo.js</code> §10 — ESOL (Delaney 2004) with its uncertainty, four drug-likeness filters reported separately, random forest with out-of-bag error	<code>test_spettrolettore</code> §4-5 — ESOL against four experimental values, mean deviation 0.68 , inside the declared ~1 log unit; forest out-of-bag R ² 0.90 on a non-linear relationship and -0.16 on pure noise; deterministic for a given seed and different for another
SCI-27	A structure without a source must not get in, not even one that is known	collected from ChEMBL, <code>tools/dati/farmaci_v187.json</code> carries each entry's identifier	Two refusals : ivermectin (<code>structure_type</code> NONE, a mixture of homologues) and semaglutide (a protein, <code>SEQ</code>). And two entries kept under the record's name rather than the query's: «morphine» → apomorphine, «levothyroxine» → liothyronine

ID	Requirement	Implementation	Verification bench
SCI-28	A predicted spectrum must say WHICH atoms produce each signal, and the deviation must be measured on molecules that did not choose the parameters	<code>bsi-prettsch.js</code> — the estimation tables transcribed in full from the source (Prettsch 4th ed.: 91 ¹³ C and 66 ¹ H rows for benzenes, 42 for ethylenes, 31 for alkanes, 24 for ¹³ C aliphatics, 4×4 steric corrections); <code>bsi-nmr.js</code> — the reasoning that applies them: ring position obtained by walking the cycle, additive scheme with a cyclic reference compound, chemical equivalence by shell environment code	<code>test_nmr</code> — 56 checks . Two separate sets: tuning 0.71 ppm over 9 molecules, validation 0.92 ppm over 22 never used for tuning, worst case 4.9 ppm , ¹ H 0.06 ppm (0.03 on the aromatics alone). The bench requires validation to stay worse than tuning: were they equal, someone would have moved a molecule between the two sets
SCI-29	Chemically equivalent atoms must give ONE signal, not one per atom	environment code built as concentric shells by distance (the HOSE-code principle), with aromatic bonds written as such and not in Kekulé form	<code>test_nmr</code> — eight molecules whose signal count is known: benzene 1 (not six), toluene 5, p-xylene 3, naphthalene 3, biphenyl 4, aspirin 9. All correct
SCI-30	From an image one can recover the SHAPE of a spectrum, not its calibration	<code>bsi-spettrolettore.js</code> §6-bis — trace extraction from the pixels column by column, background estimated from the median, empty columns interpolated	<code>test_spettrolettore</code> — on a constructed figure with Gaussians at 1715/2950/3400 cm ⁻¹ it reads 1713/2947/3403 from the pixels alone; in both directions , on a blank sheet it refuses instead of inventing a trace; and it declares that the axis scale is not in the pixels

2. AI agent requirements

ID	Requirement	Implementation	Verification bench
AI-01	A model retired by the provider must not block the assistant	<code>modelloSuggestitoDaErrore()</code> , <code>bsi_modelli_ko blacklist</code>	<code>test_404</code> — 36 checks
AI-02	A provider unreachable from the browser must be remembered and demoted	<code>bsi_prov_ko</code> , 24 h TTL, <code>providerUtilizzabili()</code>	<code>test_ko</code> (35), <code>browser_ko</code> (37)
AI-03	A temporary overload must produce a retry, not an error	<code>erroreTemporaneo()</code> , <code>TENTATIVI_TEMPORANEO</code>	<code>test_503</code> — 40 checks
AI-04	An imposed wait longer than 10 s must switch provider if a ready one exists	<code>ATTESA_TROPPO_LUNGA_MS</code>	<code>test_attesalunga</code> — 14 checks
AI-05	The provider's token ceiling must be learned from the error, not hard-coded	<code>tettoDaErrore()</code> , <code>MARGINE_TETTO</code>	<code>test_tetto</code> — 32 checks

ID	Requirement	Implementation	Verification bench
AI-06	Opaque reasoning signatures must be returned to the provider	capture and replay in <code>appendAgentTurn()</code>	<code>test_firma</code> — 31 checks
AI-07	Two overlapping submissions must not corrupt the history	<code>_turnoInCorso</code> in <code>send()</code>	<code>test_doppioinvio</code> — 7 checks
AI-08	The user must be able to measure which providers they can actually reach	<code>provaTuttiIFornitori()</code>	<code>browser_prova</code> — 29 checks
AI-09	Paid providers must not be offered as free ones	separation by <code>PROVIDERS[id].free</code>	<code>browser_prova</code> §D2
AI-10	A documented route must exist to remove CORS and rate limits	<code>proxy/spectra-proxy.js</code> + in-app guide	<code>test_guidaproxy</code> (12), <code>browser_proxy</code> (15), <code>browser_proxyui</code> (17)

3. Stability requirements

ID	Requirement	Implementation	Verification bench
STA-01	A long session must accumulate neither DOM nodes nor timers	section lifecycle management	<code>audit_stabilita</code> §1 — 92 sections × 5 rounds
STA-02	Exhausting <code>localStorage</code> must not make the app unusable	guarded writes; persistent warning in the File Manager	<code>audit_quota</code> (10), <code>test_filemanager</code> (15)
STA-03	No rejected promise may remain unhandled	systematic <code>.catch()</code>	<code>audit_promesse</code> — 22 checks, healthy network and dead network
STA-04	Corrupted saved data must not prevent startup	<code>loadJSON()</code> with fallback	<code>audit_stabilita</code> §4
STA-05	A slow network must not be treated as an absent one	<code>ATTESA_RETE_MS = 3500</code> in <code>sw.js</code>	<code>test_sw</code> §3 and §3-bis — measured 3,507 ms
STA-06	An update must not destroy work in progress	<code>BSI_SW_UPDATED</code> message, decision taken page-side	<code>test_sw</code> §4
STA-07	Browsing molecules must not exhaust the WebGL contexts	reuse of the 3D viewer at all 5 creation points	<code>test_visore3d</code> — from 7 constructions down to 1
STA-08	An unavailable IndexedDB store must be declared, not fatal	<code>dbStore()</code> over a promise; persistent warning	<code>test_filemanager</code> §2

4. Interface and accessibility requirements

ID	Requirement	Implementation	Verification bench
UI-01	Every page must open without JavaScript errors	—	<code>audit_stabilita</code> §5 — 14 pages
UI-02	Charts must be sharp on high-density screens	<code>bsiNitido()</code> on the canvas contexts	<code>audit_grafici</code> — 40 canvases
UI-03	The app must work at 390 px width	<code>auto-fit</code> grids, no fixed column	<code>audit_mobile</code> — the document's <code>scrollWidth</code> across all 92 sections at 390 px; <code>audit_stabilita</code> for errors in a phone viewport
UI-04	The user must be able to know which version they are running and force the update	"Updates" entry in the ✨ panel	<code>test_aggiorna</code> — 9 checks
UI-05	Data deletion must be selective and reversible in its choices	<code>bsiCancellaDati()</code> by group	<code>browser_reset</code> — 24 checks
UI-08	Surfaces that appear only after an action must also meet WCAG AA contrast	the six panels of the Cheminformatics section	<code>test_cheminfo</code> — the analysis is run, the panels opened in turn, 460 text elements measured, 0 defects , with the WCAG formulas rewritten inside the bench
UI-09	Opening a section must not block the page	3Dmol's first <code>render()</code> moved out of the click; the 296 synthesis figures and the 263 drug cards drawn in 8 ms slices	<code>test_fluidita</code> — 92 sections traversed , median switch 16 ms, worst 87 ms, 0 sections above 100 ms (there were 3 with the new drugs, at 159 ms); plus the opposite proofs: the WebGL canvas appears anyway, all 296 figures exist shortly after, and printing does not come out mute. For the viewer the claim is structural, not temporal : immediately after the click handler returns, the WebGL context must not exist; shortly after it must — because a millisecond threshold was measuring <code>SwiftShader</code> rather than the code (14...192 ms on identical code)
UI-10	The interface must switch among many languages, declaring how much each is translated, and carrying the writing direction with it	<code>bsi-lingue.js</code> with fourteen languages , a searchable selector, <code>dir="rtl"</code> for Arabic	<code>test_lingue</code> — 170 skeleton elements out of 170 in each of the thirteen languages , coverage computed by applying the dictionary and counting; writing direction tested in both directions; navigation categories and the search placeholder included
UI-11	The molecule must show how it holds together, and the viewer must choose the angle	<code>bsi-mol3d.js</code> : formation from dust with staggered arrival, and atom selection by click	<code>test_mol3d</code> — the canvas pixels at 130 ms and 2.3 s, the arrival order (skeleton before hydrogens) measured on each atom's progress, and the selection of 1, 2, 3 and 4 atoms

5. Security and privacy requirements

ID	Requirement	Implementation	Verification bench
SEC-01	No API key may be present in the repository	keys only in <code>localStorage</code> or in the Worker's secrets	<code>tools/verifica-sicurezza.js</code> — 307 tracked files, 8 credential shapes
SEC-02	No personal data may leave the device without an explicit action	local-first architecture, telemetry disabled	<code>audit_rete</code> — direct verification : a canary value seeded into 71 stores of the user's data, the application used across 92 sections on 6 pages, and the URL, headers and body of every request inspected. Plus <code>verifica-sicurezza</code> on the two exit mechanisms
SEC-03	Passwords must not appear in clear text in the source	SHA-256 in <code>file_manager.html</code>	<code>tools/verifica-sicurezza.js</code>
SEC-07	No credential may EVER have entered the repository, not even in a commit later fixed	no key has ever been committed; keys live in <code>localStorage</code> or in the Worker's secrets	<code>audit_storia</code> — 1,551 distinct versions of text files across 390 commits , 9 patterns tested in both directions; fails on a shallow clone, because it would measure less surface
SEC-04	No conflict marker may reach publication	—	<code>verifica_guida</code> §12 — 58 text files

SEC-03 — declared limitation. GitHub Pages serves static files: any page-side access control can be bypassed by whoever reads the source. The File Manager's protection is a deterrent, **not** a security control, and the documentation says so. Furthermore the password remained in clear text in the git history until its removal, and taking a secret out of the files does not take it out of the history: `git log -p` hands it to anyone. The only effective remedy was to change it, and that **has been done** at version `bsi-v189`. The old one remains in the history and no longer opens anything.

6. Requirements not yet verified automatically

Declared for completeness. They are covered by documentary inspection or by manual testing, and their automation is planned.

ID	Requirement	Current coverage
UI-06	Full WCAG 2.1 AA conformance	the mechanical part is automated across 13 pages and 92 sections (<code>tools/verifica-accessibilita.js</code> , 38,407 text elements: it was 19,751 before text over gradients entered the measurement, and 37,409 before the «Spectrum reader» section). Of the 1,069 contrast defects that emerged, none remain: 0 measured over those same 92 sections, and the value is recorded as a baseline the bench defends. Unlabelled fields are zero : see the box in <code>docs/09</code> §4. What remains outside is text inside SVGs and text over a real background image, counted on every run (D-09)
PERF-01	First-paint time on a low-end device	manual cross-device testing (<code>docs/02</code> §4)
SCI-11	Structures of 2 entries that are not single molecules (it was 6)	not representable : Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. The other four were closed by taking the structure from ChEMBL, see <code>docs/06</code> §2.4
PERF-02	Code coverage of the verification benches	partially measured : 49.89 % of statements over the widest path (<code>audit_copertura</code>). Whole-battery coverage and branch coverage remain unmeasured; see §8
UI-07	Behaviour on Firefox and WebKit	not verified — the battery runs on Chromium only; see §8

7. Overall coverage

Category	Requirements	Verified by a bench	Coverage
Scientific (SCI-01...27)	26	26	100 %
AI agent (AI-01...10)	10	10	100 %
Stability (STA-01...08)	8	8	100 %
Interface (UI-01...11)	9	9	100 %
Security (SEC-01...07)	5	5	100 %
Automated total	58	58	100 %
Not automated (§6)	5	0	0 %
Declared total	63	58	92 %

Coverage is computed over the requirements **declared in this document** and must not be confused with code coverage: it measures how many requirements have a bench verifying them, not how many lines are executed during the tests.

Why two totals. Reporting only the 100 % of automated requirements would be true and misleading at once: it is 100 % of what was *chosen* to be automated. The number that matters to an assessor is the second one — **54 requirements verified out of 59 declared** — and the five that are missing are listed by name in §6, not summarised into a percentage.

This table is checked by `tools/verifica-documenti.js`, which counts the identifiers actually present in the document and fails if the totals written here do not match. It is a check born of a real error: the totals said 36 over 9 scientific categories while the identifiers defined were 10, and `SCI-10` appeared twice — once as a verified requirement and once as an unverified one.

8. Declared coverage gaps

These are not defects: they are things that have **not been measured**, and that therefore cannot be asserted.

Gap	What it entails	Why it is so
Partial code coverage	Measured: 49.89 % of statements over the widest path a bench walks. It is not the coverage of the whole battery, and it is statements, not branches	Chromium's profiler collects it inside the engine, with no build and without rewriting the source: the obstacle belonged to the tool (<code>c8</code> , <code>istanbul</code>), not to the problem. Value recorded; <code>audit_copertura</code> fails if it falls
Chromium only	Behaviour on Firefox and WebKit is verified by hand, not by a bench	The battery uses <code>playwright-core</code> , which downloads a single engine. In the verification environment the CDN for the other engines answers 403 to the network policy: they cannot be installed there
No visual regression	An unintended graphical change would not be caught	Spectra are deterministic and comparable byte for byte (<code>SCI-05</code>): the comparison exists on the traces, not on the whole page
2 entries without a structure	Two entries out of 178 have no structure to show: Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. It was six	See <code>06-Scientific-Accuracy-Data-Provenance.md</code> §2.4
Performance on slow devices	First-paint time is not measured on low-end hardware	It requires physical devices; the test is manual

Document updated to version `bsi-v189`.